Lecture Notes 4

Econometrics

Heteroskedasticity

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A.Y. 2026-2027

Introduction

- Consider again the multiple linear regression model:

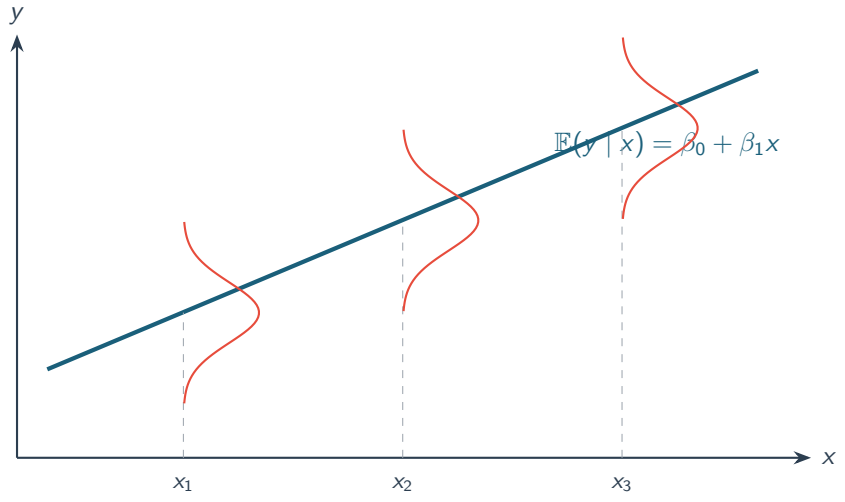
$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$$

- The homoskedasticity assumption, stated in terms of the error variance as $\text{Var}(\varepsilon|x_1, \ldots, x_k) = \sigma^2$, played no role in showing whether OLS was unbiased.
- Thus, heteroskedasticity does not cause bias in the OLS estimators of the β_j.

If heteroskedasticity does not cause bias in the OLS estimators, why did we introduce it as one of the Gauss-Markov assumptions?

What Homoskedasticity Looks Like

At each value of x, the error has a distribution around the regression line. Under **homoskedasticity** that distribution has the *same spread* everywhere:



The bell curves are **identical in width** at x_1, x_2, x_3: $\mathrm{Var}(\varepsilon \mid x) = \sigma^2$, constant for all x.

Heteroskedasticity

- Consider the model with a single independent variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- If the errors contain heteroskedasticity, then

$$\text{Var}(\varepsilon_i | x_i) = \sigma_i^2$$

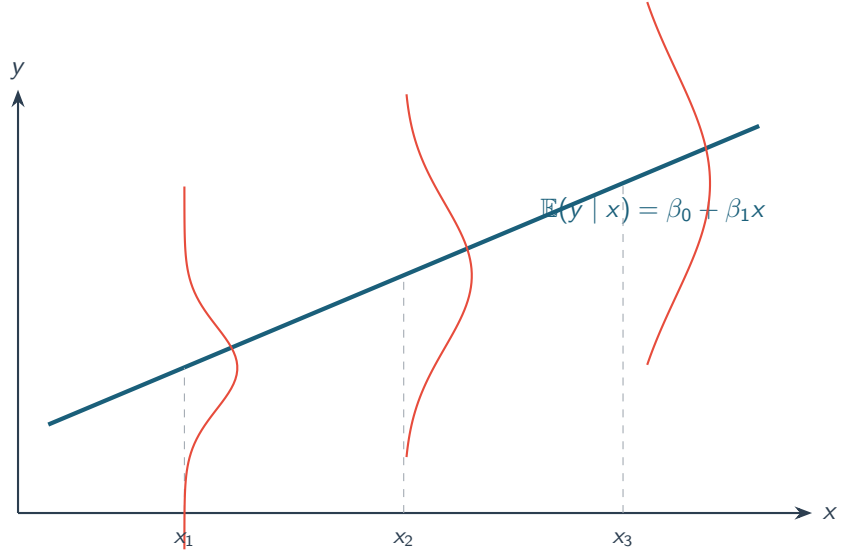
- Then, the variance of the OLS estimator is given by

$$\text{Var}(\widehat{\beta}_1) = \sigma_i \frac{\sum_{i=1}^{n} (x_i - \overline{x})^2}{TSS_x^2}$$

- When $\sigma_i^2 = \sigma^2$ for all i, this formula reduces to the usual form, σ^2 / TSS_x.
- The variance formula derived under homoskedasticity is no longer valid when heteroskedasticity is present.

What Heteroskedasticity Looks Like

Under **heteroskedasticity**, the conditional distribution has a *different spread* at different x: here it **widens** as x grows.



The bell at x_3 is much wider than at x_1: $\mathrm{Var}(\varepsilon_i \mid x_i) = \sigma_i^2$ grows with x_i.

Testing for Heteroskedasticity

- We start with the linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + \varepsilon$$

- Assume that $\mathbb{E}(\varepsilon | x_1, x_2, \ldots, x_k) = 0$, so that OLS is unbiased.
- Take the null hypothesis to be

$$H_0 : \text{Var}(\varepsilon | x_1, x_2, \ldots, x_k) = \sigma^2$$

- Thus, we assume that the ideal assumption of homoskedasticity holds, and we require the data to tell us otherwise.
- Because we are assuming that ε has a zero conditional expectation, $\text{Var}(\varepsilon | \mathbf{x}) = \mathbb{E}(\varepsilon | \mathbf{x})$, and so the null hypothesis of homoskedasticity is equivalent to

$$H_0 : \mathbb{E}(\varepsilon^2 | x_1, x_2, \ldots, x_k) = \mathbb{E}(\varepsilon^2) = \sigma^2$$

Testing for Heteroskedasticity

The Breusch-Pagan Test for Heteroskedasticity

- To test for violation of the homoskedasticity assumption, we want to test whether ε^2 is related to one or more of the explanatory variables.
- If H_0 is false, the expected value of ε^2, given the independent variables, can be virtually any function of the x_j.
- A simple approach is to assume a linear function:

$$\widehat{\varepsilon}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + \nu$$

  where ν is an error term with mean zero given the x_j.
- The null hypothesis of homoskedasticity is

$$H_0 : \delta_1 = \delta_2 = \ldots = \delta_k = 0$$

- The F statistic for the overall significance of the independent variables in explaining ε^2 can be used to test this hypothesis.
- The LM statistic for heteroskedasticity is $LM = n \times R^2_{\widehat{\varepsilon}} \sim \chi^2_k$.

Example: Heteroskedasticity in Housing Price Equations

Consider the following housing price equations:

$$\log(price) = \beta_0 + \beta_1 \log(lotsize) + \beta_2 \log(sqrft) + \beta_3 bdrms + \varepsilon$$

```
. reg lprice llotsize lsqrft bdrms
```

Source	SS	df	MS	Number of obs	=	88
				F(3, 84)	=	50.42
Model	5.15504028	3	1.71834676	Prob > F	=	0.0000
Residual	2.86256324	84	.034078134	R-squared	=	0.6430
				Adj R-squared	=	0.6302
Total	8.01760352	87	.092156362	Root MSE	=	.1846

lprice	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
llotsize	.1679667	.0382812	4.39	0.000	.0918404	.244093
lsqrft	.7002324	.0928652	7.54	0.000	.5155597	.8849051
bdrms	.0369584	.0275313	1.34	0.183	-.0177906	.0917074
_cons	-1.297042	.6512836	-1.99	0.050	-2.592191	-.001893

Example: Heteroskedasticity in Housing Price Equations

$$\widehat{\varepsilon^2} = \alpha_0 + \alpha_1 \log(lotsize) + \alpha_2 \log(sqrft) + \alpha_3 bdrms + \nu$$

```
. predict epsilon_hat, residuals

. gen epsilon_hat2 = epsilon_hat^2

. reg epsilon_hat2 llotsize lsqrft bdrms
```

Source	SS	df	MS	Number of obs	=	88
				F(3, 84)	=	1.41
Model	.022620168	3	.007540056	Prob > F	=	0.2451
Residual	.448717194	84	.005341871	R-squared	=	0.0480
				Adj R-squared	=	0.0140
Total	.471337362	87	.005417671	Root MSE	=	.07309

epsilon_hat2	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
llotsize	-.0070156	.0151563	-0.46	0.645	-.0371556	.0231244
lsqrft	-.0627368	.0367673	-1.71	0.092	-.1358526	.0103791
bdrms	.0168407	.0109002	1.54	0.126	-.0048356	.038517
_cons	.509994	.257857	1.98	0.051	-.0027829	1.022771

Testing for Heteroskedasticity

The White Test for Heteroskedasticity

- The homoskedasticity assumption, $\text{Var}(\varepsilon|x_1, \ldots, x_k) = \sigma^2$, can be replaced with the weaker assumption that the squared error, ε^2, is uncorrelated with all the independent variables (x_j), the squares of the independent variables (x_j^2), and all the cross products ($x_j x_h$ for $j \neq h$).
- White (1980): a test for heteroskedasticity that adds the squares and cross products of all the independent variables.
- The test is explicitly intended to test for forms of heteroskedasticity that invalidate the usual OLS standard errors and test statistics.
- When the model contains $k = 3$ independent variables, the White test is based on an estimation of

$$\widehat{\varepsilon^2} = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + \omega$$

- The White test is the *LM* statistic for testing the null that all of the δ_j are zero, except for the intercept.

The White Test: An Alternative

- With 6 independent variables, the White regression would generally involve 27 regressors.
- This abundance of regressors is a weakness in the pure form of the White test.
- We can preserve the spirit of the White test while conserving on degrees of freedom by using the OLS fitted values in a test for heteroskedasticity.
- Remember that the fitted values are defined, for each observation i, by

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_{i1} + \widehat{\beta}_2 x_{i2} + \ldots + \widehat{\beta}_k x_{ik}$$

- If we square the fitted values, we get a particular function of all the squares and cross products of the independent variables.

$$\widehat{\varepsilon^2} = \delta_0 + \delta_1 \widehat{y} + \delta_2 \widehat{y^2} + \nu$$

- We can use the F or LM statistic for the null hypothesis $H_0 : \delta_1 = 0, \delta_2 = 0$.

Weighted Least Squares Estimation

- The response to a finding of heteroskedasticity is to specify its form and use a weighted least squares method (WLS).
- If we have correctly specified the form of the variance, then WLS is more efficient than OLS.
- Let $\mathbf{x}$ denote all explanatory variables and assume that

$$\text{Var}(\varepsilon|\mathbf{x}) = \sigma^2 h(\mathbf{x}), \quad h(\mathbf{x}) > 0$$

where $h(\mathbf{x})$ is some function of the explanatory variables that determines the heteroskedasticity.

Weighted Least Squares Estimation

Known $h(\mathbf{x})$

- Assume the function $h(\mathbf{x})$ is known.
- For a random drawing from the population, we can write $\sigma_i^2 = \text{Var}(\varepsilon_i|\mathbf{x}_i) = \sigma^2 h(\mathbf{x}_i) = \sigma^2 h_i$.
- Take the original equation

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + \varepsilon_i$$

  which contains heteroskedastic errors, and transform it into an equation that has homoskedastic errors.
- Because h_i is just a function of $\mathbf{x}_i$, $\varepsilon_i/\sqrt{h_i}$ has a zero expected value conditional on $\mathbf{x}_i$.
- Because $\text{Var}(\varepsilon_i|\mathbf{x}_i) = \mathbb{E}(\varepsilon_i^2|\mathbf{x}_i) = \sigma^2 h_i$, the variance of $\varepsilon_i/\sqrt{h_i}$ is σ^2.
- Divide the original equation by $\sqrt{h_i}$ to get

$$\frac{y_i}{\sqrt{h_i}} = \beta_0 \frac{1}{\sqrt{h_i}} + \beta_1 \frac{x_{i1}}{\sqrt{h_i}} + \ldots + \beta_k \frac{x_{ik}}{\sqrt{h_i}} + \frac{\varepsilon_i}{\sqrt{h_i}}$$

Weighted Least Squares Estimation

Known $h(\mathbf{x})$

- We can write the transformed equation as

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \ldots + \beta_k x_{ik}^* + \varepsilon_i^*$$

- The β_j^* are examples of generalized least squares (GLS) estimators.
- The GLS estimators for correcting heteroskedasticity are called weighted least squares (WLS) estimators.

Example: Savings

- Consider the simple savings function

$$sav_i = \beta_0 + \beta_1 inc_i + \varepsilon_i, \quad \text{Var}(\varepsilon_i | inc_i) = \sigma^2 inc_i$$

- The standard deviation of ε_i conditional on inc_i, is $\sigma\sqrt{inc_i}$.
- Then, the transformed equation is

$$\frac{sav_i}{\sqrt{inc_i}} = \beta_0 \frac{1}{\sqrt{inc_i}} + \beta_1 \frac{inc_i}{\sqrt{inc_i}} + \frac{\varepsilon_i}{\sqrt{inc_i}}$$

Weighted Least Squares

Unknown $h(\mathbf{x})$

- In most cases, the exact form of heteroskedasticity is not obvious.
- We can model the function h and use the data to estimate the unknown parameters in this model.
- This results in an estimate of each h_i, denoted as $\widehat{h}_i$.
- Using $\widehat{h}_i$ instead of h_i in the GLS transformation yields an estimator called the feasible GLS (FGLS) estimator.

Feasible GLS: Modeling the Variance

A flexible and common model for the variance is exponential:

$$\text{Var}(\varepsilon \mid \mathbf{x}) = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \ldots + \delta_k x_k).$$

The exponential form guarantees $\widehat{h}_i > 0$ for any parameter estimates. Taking logs of the squared error suggests the estimating equation

$$\log(\varepsilon^2) = \alpha_0 + \delta_1 x_1 + \ldots + \delta_k x_k + e.$$

- We do not observe ε, so we replace it with the OLS residual $\widehat{\varepsilon}$.
- Regressing $\log(\widehat{\varepsilon}^2)$ on the x_j gives fitted values $\widehat{g}_i$, and we set $\widehat{h}_i = \exp(\widehat{g}_i)$.

Feasible GLS: The Procedure

1. Run OLS of y on $x_1, \ldots, x_k$ and save the residuals $\widehat{\varepsilon}_i$.
2. Compute $\log(\widehat{\varepsilon}_i^2)$.
3. Regress $\log(\widehat{\varepsilon}_i^2)$ on $x_1, \ldots, x_k$ and keep the fitted values $\widehat{g}_i$.
4. Form the estimated weights $\widehat{h}_i = \exp(\widehat{g}_i)$.
5. Estimate $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$ by WLS using weights $1/\widehat{h}_i$.

- If the variance model is correct, FGLS is more efficient than OLS and its standard errors are valid.
- Because $\widehat{h}_i$ is estimated, FGLS is no longer unbiased, but it is **consistent** and asymptotically more efficient than OLS.

What If the Variance Model Is Wrong?

FGLS relies on a *guess* for the form of $\text{Var}(\varepsilon_i \mid x)$. What if that guess $\widehat{h}_i$ is misspecified?

- **Consistency survives.** FGLS remains consistent for β even when the variance model is wrong — it is still a weighted regression, and the weights do not bias the coefficient estimates.
- **Efficiency is lost.** The whole point of FGLS was efficiency. If the weights are wrong, that gain disappears — FGLS may even be *less* efficient than plain OLS.
- **Standard errors become invalid.** The FGLS standard errors are correct only if the variance model is right. Under misspecification they are unreliable — the same problem we started with.

So FGLS trades one strong assumption for another: to gain efficiency, we must *correctly specify the entire variance function* — a demanding requirement.

A Simpler, More Robust Idea

Instead of modeling the variance, leave OLS alone and just **correct its standard errors** — no guess about the form of heteroskedasticity required.

Heteroskedasticity-Robust Standard Errors

The key insight (White, 1980): we do not need to *know* each σ_i^2 to estimate the variance of $\widehat{\beta}_1$ — the squared residuals $\widehat{\varepsilon}_i^2$ already carry that information.

- Recall the true variance under heteroskedasticity:

$$\text{Var}(\widehat{\beta}_1) = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2 \sigma_i^2}{TSS_x^2}.$$

- Replace each unknown σ_i^2 with the squared OLS residual $\widehat{\varepsilon}_i^2$:

$$\widehat{\text{Var}}(\widehat{\beta}_1) = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2 \widehat{\varepsilon}_i^2}{TSS_x^2}.$$

- The square root is the **robust** (or White) standard error.
- Valid in large samples whether or not errors are homoskedastic — so it has become the default in modern applied work.
- Keeps the OLS estimates unchanged; only the standard errors (and hence t's and CIs) are corrected.

Example: Financial Wealth Equation

Consider the following regression model

$$nettfa = \beta_0 + \beta_1 e401k + \beta_2 male + \beta_3 (age - 25)^2 + \beta_4 inc + \varepsilon$$

where:

- *nettfa* is net total financial wealth,
- $e401k$ is an indicator for whether the person is eligible for a 401(k) pension plan,
- *male* is a indicator for whether the individual is a male,
- $(age - 25)^2$ is a quadratic function in age,
- *inc* is income.

Example: Financial Wealth Equation

Estimation Results

```
. reg nettfa e401k male age25_squared inc
```

Source	SS	df	MS		
Model	583841.332	4	145960.333	Prob > F	0.0000
Residual	3982123.72	2,012	1979.18674	R-squared	0.1279
				Adj R-squared	0.1261
Total	4565965.05	2,016	2264.86361	Root MSE	44.488

	Number of obs	=	2,017
	F(4, 2012)	=	73.75

nettfa	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
e401k	6.886223	2.123275	3.24	0.001	2.722176	11.05027
male	2.477927	2.047776	1.21	0.226	-1.538056	6.49391
age25_squared	.0251267	.0025934	9.69	0.000	.0200407	.0302127
inc	.7705833	.061452	12.54	0.000	.6500671	.8910995
_cons	-20.98499	2.472022	-8.49	0.000	-25.83298	-16.137

Example: Financial Wealth Equation

Testing for Heteroskedasticity – Breusch-Pagan Test

```
. reg res2 e401k male age25_squared inc
```

Source	SS	df	MS		
Model	1.7483e+10	4	4.3709e+09	Prob > F	0.0034
Residual	2.2271e+12	2,012	1.1069e+09	R-squared	0.0078
				Adj R-squared	0.0058
Total	2.2446e+12	2,016	1.1134e+09	Root MSE	33270

Number of obs	=	2,017				
F(4, 2012)	=	3.95				
Prob > F	=	0.0034				
R-squared	=	0.0078				
Adj R-squared	=	0.0058				
Root MSE	=	33270				

res2	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
e401k	1164.827	1587.888	0.73	0.463	-1949.251	4278.904
male	2331.253	1531.427	1.52	0.128	-672.0949	5334.601
age25_squared	4.848656	1.93946	2.50	0.012	1.045095	8.652217
inc	112.3581	45.9568	2.44	0.015	22.2302	202.486
_cons	-4573.552	1848.699	-2.47	0.013	-8199.116	-947.9886

Example: Financial Wealth Equation

Testing for Heteroskedasticity – White Test

Source	SS	df	MS	Number of obs	=	2,017
				F(12, 2004)	=	2.79
Model	3.6826e+10	12	3.0688e+09	Prob > F	=	0.0009
Residual	2.2078e+12	2,004	1.1017e+09	R-squared	=	0.0164
				Adj R-squared	=	0.0105
Total	2.2446e+12	2,016	1.1134e+09	Root MSE	=	33192

res2	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
e401k	-467.2759	3947.069	-0.12	0.906	-8208.065	7273.513
male	-4575.786	3485.98	-1.31	0.189	-11412.31	2260.74
age25_squared	-19.7117	7.71872	-2.55	0.011	-34.84925	-4.574141
inc	-101.0215	159.5468	-0.63	0.527	-413.9164	211.8733
e401k2	0	(omitted)				
male2	0	(omitted)				
age25_squared2	.0103912	.0049659	2.09	0.037	.0006524	.0201301
inc2	-.5738566	1.5554	-0.37	0.712	-3.624228	2.476514
e401k_male	863.1458	3300.001	0.26	0.794	-5608.646	7334.938
e401k_age25	-4.641063	4.380578	-1.06	0.290	-13.23203	3.949902
e401k_inc	93.66378	94.86656	0.99	0.324	-92.38362	279.7112
male_age25	5.695133	4.080879	1.40	0.163	-2.308076	13.69834
male_inc	157.266	103.3351	1.52	0.128	-45.38957	359.9215
age25_inc	.4102446	.1394001	2.94	0.003	.1368604	.6836288
_cons	3860.21	3747.109	1.03	0.303	-3488.426	11208.85

Example: Financial Wealth Equation

Testing for Heteroskedasticity – Alternative White Test

```
. reg res2 yhat yhat2
```

Source	SS	df	MS		
Model	1.9435e+10	2	9.7175e+09	Number of obs	= 2,017
Residual	2.2252e+12	2,014	1.1048e+09	F(2, 2014)	= 8.80
				Prob > F	= 0.0002
Total	2.2446e+12	2,016	1.1134e+09	R-squared	= 0.0087
				Adj R-squared	= 0.0077
				Root MSE	= 33239

res2	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yhat	21.59607	88.61904	0.24	0.807	-152.1985	195.3906
yhat2	3.198666	1.735553	1.84	0.065	-.2050011	6.602334
_cons	163.6069	974.0734	0.17	0.867	-1746.69	2073.904

Example: Financial Wealth Equation

Weighted Least Squares

Assume that $\text{Var}(\varepsilon) = \sigma^2 inc$. Thus, to obtain homoskedastic errors, divide the model by $\sqrt{inc_i}$:

$$\frac{nettfa_i}{\sqrt{inc_i}} = \beta_0 \frac{1}{\sqrt{inc_i}} + \beta_1 \frac{e401k_i}{\sqrt{inc_i}} + \beta_2 \frac{male_i}{\sqrt{inc_i}} + \beta_3 \frac{(age_i - 25)^2}{\sqrt{inc_i}} + \beta_4 \sqrt{inc_i} + \varepsilon_i^*$$

where

$$\text{Var}(\varepsilon_i^* \mid X_i) = \sigma^2$$

Example: Financial Wealth Equation

Weighted Least Squares – Results

Financial Wealth Equation

	(1) OLS	(2) WLS
e401k	6.886***	5.188***
	(2.123)	(1.703)
male	2.478	1.841
	(2.048)	(1.564)
age25_squa~d	0.0251***	0.0175***
	(0.00259)	(0.00193)
inc	0.771***	0.740***
	(0.0615)	(0.0643)
_cons	-20.98***	-16.70***
	(2.472)	(1.958)
Observations	2017	2017
R-squared	0.128	0.112

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Example: Financial Wealth Equation

Feasible GLS

Suppose now that $\text{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2 h(\mathbf{x}_i)$, but the function $h(\mathbf{x}_i)$ is unknown.
Let's model the variance as

$$\text{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2 \exp\left(\delta_0 + \delta_1 e401k_i + \delta_2 male_i + \delta_3 (age_i - 25)^2 + \delta_4 inc_i\right)$$

Example: Financial Wealth Equation

Feasible GLS

Financial Wealth Equation

	(1) OLS	(2) WLS	(3) FGLS
e401k	6.886***	5.188***	4.171***
	(2.123)	(1.703)	(1.195)
male	2.478	1.841	1.430
	(2.048)	(1.564)	(1.010)
age25_squa~d	0.0251***	0.0175***	0.0149***
	(0.00259)	(0.00193)	(0.00252)
inc	0.771***	0.740***	0.443***
	(0.0615)	(0.0643)	(0.0567)
_cons	-20.98***	-16.70***	-9.266***
	(2.472)	(1.958)	(1.423)
Observations	2017	2017	2017
R-squared	0.128	0.112	0.0624

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Example: Financial Wealth Equation

Heteroskedasticity-Robust Standard Errors

Financial Wealth Equation

	(1) Default SE	(2) Robust SE
e401k	6.886***	6.886***
	(2.123)	(2.287)
male	2.478	2.478
	(2.048)	(2.058)
age25_squ~d	0.0251***	0.0251***
	(0.00259)	(0.00434)
inc	0.771***	0.771***
	(0.0615)	(0.0996)
_cons	-20.98***	-20.98***
	(2.472)	(3.495)
Observations	2017	2017
R-squared	0.128	0.128

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

- OLS coefficient estimates are unchanged.
- Only the estimated standard errors are adjusted.
- Robust standard errors remain valid under unknown forms of heteroskedasticity.

Cluster-Robust Standard Errors

White standard errors fix heteroskedasticity but still assume errors are **uncorrelated across observations**. That fails when the data fall into **groups** (clusters) whose errors move together.

- **Examples.** Students within the same school; workers within the same firm; repeated observations on the same individual over time (panel data).
- Within a cluster, the ε_i may be correlated: $\text{Cov}(\varepsilon_i, \varepsilon_j) \neq 0$ for i, j in the same group. This violates the no-serial-correlation assumption.
- Ignoring it makes standard errors **too small** — the sample has less independent information than n observations suggest, so t-statistics are overstated.

The Idea

Allow *arbitrary* correlation **within** each cluster, while assuming independence **across** clusters.

Cluster-Robust Standard Errors

The Estimator

Suppose the n observations are partitioned into G clusters, indexed $g = 1, \ldots, G$. The cluster-robust variance estimator is

$$\widehat{\mathrm{Var}}(\widehat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^{G} \mathbf{X}_g' \, \widehat{\varepsilon}_g \, \widehat{\varepsilon}_g' \, \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1},$$

where $\mathbf{X}_g$ and $\widehat{\varepsilon}_g$ collect the rows and residuals belonging to cluster g.

- It is a **sandwich estimator**, like White's — but the "meat" sums over *clusters*, using the full residual vector $\widehat{\varepsilon}_g$ within each. This captures the within-cluster covariances.
- If every cluster has size one ($G = n$), it collapses back to the **White** heteroskedasticity-robust estimator.
- **Caveat.** The approximation relies on *many* clusters. With few clusters (a common rule of thumb is fewer than ∼30–50), cluster-robust standard errors can be unreliable.

References

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